INDUSENSE SYNTHETIC MAINTENANCE MANUAL
DOCUMENT ID: SYNTH-MAN-BRG-024
TITLE: PRONOSTIA Roller Bearing Degradation & Lubrication Standards
STATUS: SYNTHETIC DATASET -- RESEARCH FRAMEWORK ONLY

SECTION 3: ROLLER BEARING VIBRATION SIGNATURES
Outer race spalling manifests through repetitive shock pulses at Ball Pass Frequency Outer Race:
BPFO = (N_balls / 2) * (1 - (d_roller / D_pitch) * cos(contact_angle)) * shaft_frequency.
For benchmark speed 1800 RPM, BPFO modulates at 108.4 Hz.

SECTION 6.1: DEGRADATION THRESHOLDS & STAGE-3 SPALLING
Condition Criteria:
- Horizontal Vibration Kurtosis > 4.5
- Envelope Spectrum Modulation at 108 Hz > 14 dB
- Crest Factor > 4.0

ACTION PROTOCOL:
Denotes Stage-3 subsurface fatigue spalling. Immediately inspect grease contamination level.
If metal particulates are identified on magnetic drain plug, schedule bearing cartridge replacement.